Short extrapolation and growing zero jet bands for a theta prime Gamma form

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Abstract

We prove a coefficient independent extrapolation inequality for exponential polynomials whose exponents lie in a vertical strip, including every polynomial multiplicity prefix. A weighted elimination identity and an orthonormal stable Volterra basis give constants independent of frequency separation and confluence. Applied to the full theta weight, the inequality localizes ordinary L^2 mass at the scale $e^{2r} = 100N/\sqrt{\log N}$ for prefix budget $N \geq 4096$. Consequently the complete prime power form has norm at most $12\sqrt{\lceil 100N/\sqrt{\log N}\rceil + 1}$ relative to the ordinary L^2 norm. A direct derivative argument and the complete Gamma multiplier then prove $q_0[u] = (1+o(1))\log T\,\|u\|_2^2$ uniformly on the entire actual zeta zero jet space in $[T, T+L(T)]$ whenever $L(T) = o(\log T\sqrt{\log\log T})$. All zeros in the band, all allowed multiplicity derivatives and all internal couplings are retained. We also give a conservative positive result at a small fixed multiple of the boundary scale and an exact inverse retaining Schur consequence. No Riemann hypothesis or prime number theorem is assumed. No zeros are enumerated here; a published explicit zero counting theorem is invoked. These one band results do not establish nonnegativity on arbitrary unions or the Riemann hypothesis.

1 The form and the main conclusions

All logarithms are natural. Norms with subscript 2 are ordinary, unnormalized Lebesgue L^2 norms, on the interval indicated or on $\mathbb{R}$. Inner products are linear in the first argument. For $r = |x|$ define the positive even weight

$$W(x) = \frac{1}{1+e^{-r}} \sum_{j=1}^{\infty} (4\tau_j^2 - 6\tau_j) e^{-\tau_j}, \qquad \tau_j = \pi j^2 e^{2r}. \tag{1}$$

Let Λ be the von Mangoldt function and put

$$\begin{aligned} \rho(s) &= \frac{e^{-s/2}}{1-e^{-2s}}, \qquad a_n = \frac{\Lambda(n)}{\sqrt{n}}, \qquad c_\Gamma = \log\pi - \psi(1/4), \quad \psi = \Gamma'/\Gamma, \\ t_\Gamma[u] &= \int_0^\infty \rho(s)\|u(\cdot+s) - u\|_2^2\, ds, \\ p[u] &= 2\,\mathrm{Re}\sum_{n=2}^{\infty} a_n \int_{\mathbb{R}} u(x+\log n)\overline{u(x)}\, dx, \\ q_0[u] &= t_\Gamma[u] - c_\Gamma\|u\|_2^2 - p[u]. \end{aligned} \tag{2}$$

The sum runs over all integers, so it includes every prime power with its correct von Mangoldt coefficient and both translation orientations. We use these full forms, not finite heads. They are well defined on every $u = WP$ considered below. Such functions belong to $L^2(\mathbb{R}, dx/W) \cap H^1(\mathbb{R})$; the estimates below prove absolute convergence of the prime sum and finiteness of the Gamma integral.

An exponential polynomial has prefix budget at most N if it has a representation

$$P(x) = \sum_{j=1}^{J} p_j(x) e^{z_j x}, \qquad \deg p_j < m_j, \qquad \sum_j m_j \le N. \tag{3}$$

Equal z_j may be merged. A selected degree k costs its entire prefix of length $k+1$ even if some coefficients vanish. No coefficient norm, separation condition or inverse Vandermonde estimate is imposed.

To state the actual zero consequence, write

$$\xi(s) = \tfrac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \Xi(w) = \xi(1/2 + iw).$$

For each zero w of Ξ with $\operatorname{Re} w > 0$, let $\nu(w)$ be its order and set

$$J_{w,k}(x) = W(x)\, \partial_z^k \cos(zx)\Big|_{z=w}, \qquad 0 \le k < \nu(w). \tag{4}$$

Let $V_{T,L}$ be the complex span of all these vectors with $T \le \operatorname{Re} w \le T + L$, including every prefix in (4). Zeros off the critical line are included: the classical critical strip only gives $|\operatorname{Im} w| < 1/2$ [5, Sections 25.4 and 25.10].

Theorem 1.1 (Entire growing actual bands)**.** *Let $L(T) \ge 0$ satisfy*

$$L(T) = o\big((\log T)\sqrt{\log\log T}\big) \qquad (T \longrightarrow \infty). \tag{5}$$

There is a scalar $\varepsilon_T \to 0$, independent of the coefficient vector in $V_{T,L(T)}$, such that

$$(1 - \varepsilon_T)\log T\, \|u\|_2^2 \le q_0[u] \le (1 + \varepsilon_T)\log T\, \|u\|_2^2, \qquad u \in V_{T,L(T)}. \tag{6}$$

In particular, this holds for $L(T) = (\log T)(\log\log T)^{1/4}$. The norm is that of the entire represented sum, with all correlations inside the band. Empty spaces cause no difficulty.

The analytic estimates hold for arbitrary exponents in the strip. Zeta specific information enters only through the strip location and the zero count with multiplicity.

Corollary 1.2 (A fixed small boundary coefficient)**.** *For every fixed $d > 0$,*

$$\liminf_{T\to\infty} \inf_{0 \ne u \in V_{T,d(\log T)\sqrt{\log\log T}}} \frac{q_0[u]}{\log T\, \|u\|_2^2} \ge 1 - \frac{120\sqrt{d}}{2^{3/4}\sqrt{\pi}}. \tag{7}$$

Thus every fixed $0 < d < \pi\sqrt{2}/7200$ gives eventual strict positivity on the whole band. This constant is conservative; no assertion is made at equality.

1.1 Relation to prior methods and the scope of the result

Zhang's real frequency sparse Fourier extrapolation theorem [1, Theorem 1.1 and Proposition 3.7] supplies the starting methodological comparison. His weighted frequency elimination and energy concentration argument is the model for Lemma 2.1. The stated frequency class in that paper has real Fourier frequencies, equivalently purely imaginary exponential roots, and does not itself state the strip and polynomial prefix conclusion needed here. We give this extension directly, and combine it with an orthonormal Volterra construction. At zero strip width our explicit bound recovers the constant in Zhang's Proposition 3.7 after translation and rescaling; see Remark 3.3. Erdélyi's work [2] provides an earlier general setting for coefficient independent inequalities for exponential sums. Our derivative estimate is proved below from the explicit basis, rather than being inferred from a compactness argument.

The theta form normalization and the carrier comparison were used in preceding stages of the author's research. This paper reproduces their definitions and the estimates it needs; no earlier manuscript of the author is an external hypothesis. The new localization scale improves the $O(\sqrt{N})$ prime loss available from a linear physical cutoff to $O(\sqrt{N}/(\log N)^{1/4})$. This is the gain that allows (5). We do not assert optimality of that scale or its constants.

No local positivity result can simply be summed across arbitrary bands: the mixed terms of q_0 remain. Section 9 retains the full inverse response within a union already covered by the whole band estimate. It does not provide an exhaustive gluing theorem. In particular, (6) is an ordinary L^2 comparison on growing finite dimensional spaces, not a uniform gap in $L^2(dx/W)$. Neither global nonnegativity nor the Riemann hypothesis follows.

2 A finite interval energy comparison

Lemma 2.1 (Weighted half line concentration)**.** *Suppose P has prefix budget at most $N \geq 1$ and $|\operatorname{Re} z_j| \leq a$, where $a \geq 0$. For $\ell > 0$ set*

$$\alpha = 8N/\ell + 2a, \qquad w(t) = e^{\alpha(t-\ell)}.$$

Then

$$E(P) := \int_{-\infty}^{\ell} w(t)|P(t)|^2\,dt \leq 2\|P\|^2_{L^2[0,\ell]}. \tag{8}$$

The reflected version is

$$\int_0^{\infty} e^{-\alpha t}|P(t)|^2\,dt \leq 2\|P\|^2_{L^2[0,\ell]}. \tag{9}$$

Proof. At an exponent z of a current exponential polynomial f, put $c = \alpha + 2\operatorname{Re} z > 0$ and define

$$b(t) = c\int_0^{\infty} e^{-(c-z)s} f(t-s)\,ds, \qquad h = f - b.$$

The denominator associated with another root z_j has real part $\alpha + \operatorname{Re} z + \operatorname{Re} z_j \geq \alpha - 2a > 0$. All integrals therefore converge, including polynomial factors. For the selected root, convolution sends $e^{zt}p(t)$ to $e^{zt}(p(t) - p'(t)/c + \cdots)$, so h decreases that root's polynomial prefix by at least one. Other prefixes do not increase. Successive elimination gives $f_0 = P$, $f_m = 0$ with $m \leq N$ and $b_j = f_j - f_{j+1}$, $0 \leq j < m$.

Differentiation gives $b' = zb + ch$ and the exact identity

$$(w|b|^2)' = c\,w(|f|^2 - |h|^2). \tag{10}$$

The boundary at $-\infty$ vanishes because $\alpha > 2a$. Integration yields

$$E(f_j) - E(f_{j+1}) = |b_j(\ell)|^2/c_j \geq 0.$$

Multiplication by $\ell - t$ and integration by parts yield

$$\int_{-\infty}^{\ell} (\ell - t)w(t)|P(t)|^2\,dt = \sum_{j=0}^{m-1} \frac{E(b_j)}{c_j} \leq \frac{4m}{\alpha - 2a}E(P) \leq \frac{\ell}{2}E(P).$$

Here $E(b_j) \leq 2E(f_j) + 2E(f_{j+1}) \leq 4E(P)$. For $t < 0$, $\ell - t \geq \ell$, so the weighted energy there is at most $E(P)/2$. On $[0, \ell]$, $w \leq 1$. This proves (8). Apply it to $P(\ell - t)$ and change variables to obtain (9). The argument includes repeated roots directly, without taking a limit of a singular coefficient Gram matrix. □

3 The stable Volterra basis and short extrapolation

The orthonormal construction below is the classical right half plane Takenaka–Malmquist system, written as a product of stable Volterra all pass operators. Zhang [1, Appendix A, especially Remark A.7] discusses the all pass formulation. We include its proof, endpoint identities and continuation estimate to cover varying positive real parts and every repeated root prefix explicitly. The construction itself is not claimed as new.

Lemma 3.1 (An orthonormal stable basis)**.** *Let $\zeta_1, \dots, \zeta_m$ have positive real parts b_j, with repetitions allowed. On $L^2(0, \infty)$ define*

$$(T_\zeta h)(t) = h(t) - 2\operatorname{Re}\zeta \int_0^t e^{-\zeta(t-v)}h(v)\,dv. \tag{11}$$

The functions

$$e_j(t) = \sqrt{2b_j}\,e^{-\zeta_j t}, \qquad \phi_j = T_{\zeta_1} \cdots T_{\zeta_{j-1}} e_j \tag{12}$$

are an orthonormal basis of the exponential polynomial space with root list $-\zeta_1, \dots, -\zeta_m$, including its repeated root prefixes. If $0 < b_j \leq B$, then for $d \geq 0$,

$$|\phi_j(-d)| \leq \sqrt{2B}\exp\bigl(Bd + 2\sqrt{2B(j-1)d}\bigr). \tag{13}$$

Also

$$\phi_j(0) = \sqrt{2b_j}, \qquad \phi_j'(0) = -\Bigl(\zeta_j + 2\sum_{k<j} b_k\Bigr)\sqrt{2b_j}. \tag{14}$$

Proof. Extend h by zero to the negative half line. The full line Fourier multiplier of (11) is $(i\xi - \overline{\zeta})/(i\xi + \zeta)$, of modulus one for real ξ. Causality makes it an isometry on the half line. Direct Fubini gives $\langle T_\zeta h, e^{-\zeta t}\rangle_{L^2(0,\infty)} = 0$: the inner convolution integral has denominator $\zeta + \overline{\zeta} = 2\operatorname{Re}\zeta$. These assertions first hold for compactly supported functions and extend by boundedness. For $i < j$, cancel the first $i - 1$ common isometries in the inner product of ϕ_i and ϕ_j, and use this orthogonality for T_{ζ_i}. Each ϕ_j belongs to the prefix space of the first j roots. That space has dimension j, including repetitions. Thus they are a basis.

Their unique exponential polynomial continuations to negative real t satisfy, on reversing the finite integral in (11),

$$|(T_\zeta h)(-d)| \leq |h(-d)| + 2B\int_0^d e^{B(d-v)}|h(-v)|\,dv.$$

After multiplying by e^{-Bd}, iterate $I + 2BV$, where $(Vg)(d) = \int_0^d g(v)\,dv$. The resulting majorant is

$$\sqrt{2B}\, e^{Bd} \sum_{k=0}^{j-1} \binom{j-1}{k} \frac{(2Bd)^k}{k!}.$$

Use $\binom{j-1}{k} \le (j-1)^k/k!$ and $\sum_{k\ge 0} y^k/(k!)^2 \le e^{2\sqrt{y}}$ for $y \ge 0$. The latter follows from the even terms of the exponential series, since $\binom{2k}{k} \le 4^k$. This proves (13). At zero, $T_\zeta h(0) = h(0)$ and $(T_\zeta h)'(0) = h'(0) - 2\,\mathrm{Re}\,\zeta\, h(0)$. Iteration proves (14). □

Theorem 3.2 (Complex strip short extrapolation)**.** *Suppose* (3) *holds,* $|\mathrm{Re}\, z_j| \le a$, $\ell > 0$, *and* $N \ge 1$. *Put*

$$\alpha = 8N/\ell + 2a, \qquad C = \alpha + 2a = 8N/\ell + 4a. \tag{15}$$

For every $d \ge 0$,

$$|P(\ell + d)| \le \sqrt{2NC}\exp(ad + 2\sqrt{NCd})\|P\|_{L^2[0,\ell]}. \tag{16}$$

Constants are independent of imaginary parts, coefficient conditioning, collisions and multiplicity distribution.

Proof. Set $H(u) = e^{-\alpha u/2}P(\ell - u)$, $u \ge 0$. Its stable roots are $-\zeta_j$, with $\zeta_j = \alpha/2 + z_j$ and $0 < \mathrm{Re}\,\zeta_j \le C/2$. Lemma 2.1 gives $\|H\|_{L^2(0,\infty)} \le \sqrt{2}\|P\|_{L^2[0,\ell]}$. Expanding in Lemma 3.1 and applying Cauchy–Schwarz gives

$$|H(-d)| \le \sqrt{NC}\, e^{Cd/2 + 2\sqrt{NCd}}\|H\|_{L^2(0,\infty)}.$$

Since $P(\ell + d) = e^{-\alpha d/2}H(-d)$ and $C - \alpha = 2a$, (16) follows. A shorter list only decreases the displayed upper bounds. □

Remark 3.3. At $a = 0$, $\ell = 2$, the prefactor in (16) is $2\sqrt{2}\,N$ and the exponential factor is $e^{4N\sqrt{d}}$. Translation to $[-1, 1]$ recovers Zhang's Proposition 3.7 [1]. The strip and full prefix cases are separately proved extensions. No sharpness at large extrapolation distances is asserted.

4 The full theta tail

Lemma 4.1 (Theta envelopes)**.** *For* $r = |x|$,

$$c_W e^{4r - \pi e^{2r}} \le W(x) \le C_4 e^{4r - \pi e^{2r}}, \qquad c_W = 2\pi^2 - 3\pi > 9, \quad C_4 < 42. \tag{17}$$

The weight is locally absolutely continuous and, almost everywhere,

$$|(\log W)'(x)| \le 112 e^{2|x|}. \tag{18}$$

Proof. Every term in (1) is positive. The $j = 1$ term and $1 \le 1 + e^{-r} \le 2$ give the lower bound. For the upper bound take $C_4 = 4\pi^2 \sum_{j\ge 1} j^4 e^{-\pi(j^2-1)}$. The first term beyond $j = 1$ is less than $1/32$, using $\pi > 3$, $e > 2$. Consecutive tail terms have ratio at most $(3/2)^4 e^{-15} < 1/2$. Thus the full sum is less than $17/16$, and $\pi < 22/7$ gives $C_4 < 2057/49 < 42$. Also $c_W > 9$ follows from $\pi > 3$.

For $r > 0$, differentiation is locally uniform. Write $g(r) = (1 + e^{-r})^{-1}$, so $0 < g \le 1$, $0 \le g' \le 1$. The absolute derivative of a summand times g is at most

$$(8\tau_j^3 + 32\tau_j^2 + 18\tau_j)e^{-\tau_j} \le 21\tau_j^3 e^{-\tau_j}, \qquad \tau_j \ge \pi > 3.$$

Hence $|W'| \le C_6 e^{6r-\pi e^{2r}}$, where $C_6 = 21\pi^3 \sum_{j\ge1} j^6 e^{-\pi(j^2-1)} < 1000$. Indeed the first tail term is less than $1/8$, consecutive tail ratios are at most $(3/2)^6 e^{-15} < 1/2$, and $21(22/7)^3(5/4) < 1000$. The same bound holds almost everywhere for the continuous even continuation, giving local absolute continuity at zero. Divide by the lower bound and use $1000/9 < 112$. □

Theorem 4.2 (An explicit sublinear mass cutoff)**.** *Suppose P has prefix budget at most $N \ge 4096$ and $|\operatorname{Re} z_j| \le 1/2$. Set*

$$s = \log N, \quad M_N = \frac{100N}{\sqrt{s}}, \quad R_N = \tfrac{1}{2}\log M_N, \quad U = \|WP\|_2. \tag{19}$$

For every $r \ge R_N$, writing $v = e^{2r}$,

$$|W(r)P(\pm r)|^2 \le e^{-4v}U^2, \tag{20}$$

$$\|\mathbf{1}_{\{|x|>r\}}WP\|_2^2 \le \frac{e^{-4v}}{4v}U^2 \le e^{-4v}U^2. \tag{21}$$

Moreover,

$$\int_{\mathbb{R}} e^{4|x|}|W(x)P(x)|^2\,dx \le (M_N^2+1)U^2. \tag{22}$$

Proof. The theta envelope gives square integrability before normalization. Assume $U > 0$. For $r > 1$ apply Theorem 3.2 with $a = 1/2$, $\ell = r-1$ and $d = 1$, and reflect for the negative ray. The lower comparison function is decreasing on $x \ge 0$, so

$$\|P\|_{L^2[0,r-1]} \le c_W^{-1} e^{-4(r-1)+\pi e^{2(r-1)}} U.$$

Multiplication by the upper envelope gives

$$\frac{|W(r)P(\pm r)|^2}{U^2} \le \left(\frac{C_4}{c_W}\right)^2 e^8 (2NC_r)\exp\bigl(1+4\sqrt{NC_r}-2\pi(1-e^{-2})e^{2r}\bigr), \quad C_r = \frac{8N}{r-1}+2. \tag{23}$$

Here $s > 8$, $R_N \ge s/4$ and $R_N - 1 \ge s/8$. On the entire ray $r \ge R_N$, $C_r \le 64N/s + 2 \le 66N/s$. The logarithm of the prefactor in (23), including its factor e^1, is at most

$$\log 25 + 9 + \log 132 + 2s - \log s < 18 + 2s \le 3N/\sqrt{s}.$$

For the last whole ray inequality, $(18+2s)\sqrt{s}/e^s$ is decreasing for $s \ge 8$ and is less than 3 at 8. Also $4\sqrt{NC_r} < 33N/\sqrt{s}$ and $2\pi(1-e^{-2}) > 5$. Since $e^{2r} \ge 100N/\sqrt{s}$,

$$\log \frac{|W(r)P(\pm r)|^2}{U^2} \le 36N/\sqrt{s} - 5e^{2r} \le -4e^{2r}.$$

This proves (20) everywhere on the ray. Integrating both rays gives

$$2\int_r^\infty e^{-4e^{2x}}\,dx = \int_v^\infty \frac{e^{-4t}}{t}\,dt \le \frac{e^{-4v}}{4v},$$

proving (21). Finally the layer cake identity gives

$$\int e^{4|x|}|WP|^2 \le M_N^2U^2 + 4\int_{R_N}^\infty e^{4r}\|\mathbf{1}_{\{|x|>r\}}WP\|_2^2\,dr \le \left(M_N^2 + \frac{e^{-4M_N}}{8}\right)U^2.$$

This implies (22). □

5 Every prime power and the complete prime estimate

Lemma 5.1 (Elementary prime head)**.** *For every $X \geq 1$,*

$$\sum_{2 \leq n \leq X} \frac{\Lambda(n)}{\sqrt{n}} \leq 6\sqrt{X}, \qquad |p_{n \leq X}[u]| \leq 12\sqrt{X}\,\|u\|_2^2. \tag{24}$$

Also $0 \leq a_n \leq 1$.

Proof. Write $\Psi(x) = \sum_{n \leq x} \Lambda(n)$. The factorial valuation formula gives $\Psi(2m) - \Psi(m) \leq \log \binom{2m}{m} \leq 2m \log 2$. Every prime power in $(m, 2m]$ contributes one to the corresponding valuation, whose other summands are nonnegative. At powers of two, summation gives $\Psi(2^k) \leq (2^{k+1} - 2)\log 2$. For $2^{k-1} < x \leq 2^k$ it follows that $\Psi(x) < 4(\log 2)x < 3x$, with the same weak bound at endpoints. Partial summation gives

$$\sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n}} = \frac{\Psi(X)}{\sqrt{X}} + \frac{1}{2}\int_1^X \Psi(t) t^{-3/2}\, dt \leq 6\sqrt{X}.$$

Each translation correlation is at most $\|u\|_2^2$ in modulus, which proves the two orientation head bound. Finally $a_n \leq (\log n)/\sqrt{n} \leq 2/e < 1$. □

Theorem 5.2 (Complete strip prime cap)**.** *Under the hypotheses of Theorem 4.2, the full prime series is absolutely convergent and*

$$|p[WP]| \leq P_N \|WP\|_2^2, \qquad P_N = 12\sqrt{\left\lceil \frac{100N}{\sqrt{\log N}} \right\rceil} + 1. \tag{25}$$

Thus $P_N = O(\sqrt{N}/(\log N)^{1/4})$.

Proof. For $C_u(s) = \int u(x+s)\overline{u(x)}\, dx$, splitting into $|x| > s/2$ and its complement yields

$$|C_u(s)| \leq 2\|u\|_2 \|\mathbf{1}_{\{|x| > s/2\}} u\|_2, \qquad s > 0. \tag{26}$$

On the complement $|x + s| \geq s/2$, justifying the second term after a change of variables. Set $Y = \lceil M_N \rceil$ and use Lemma 5.1 for the head. For every integer $n > Y$, (21) at $r = (\log n)/2$ gives $|C_u(\log n)| \leq 2e^{-2n}\|u\|_2^2$. The remaining series, with both orientations, costs at most $4\sum_{n > Y} e^{-2n}\|u\|_2^2 < \|u\|_2^2$. This proves absolute convergence and (25). No cancellation of carrier phases is used. □

6 A coefficient independent derivative estimate

The same basis gives an explicit polynomial derivative bound, useful because both offset width and prefix count grow.

Lemma 6.1 (A local derivative bound)**.** *Suppose Q has prefix budget at most $N \geq 1$, with $|\operatorname{Re} z_j| \leq 1/2$ and $|\operatorname{Im} z_j| \leq N$. For every $x \in \mathbb{R}$ and $0 < \delta \leq 1$,*

$$|Q'(x)| \leq 100 N^3 \delta^{-3/2} \|Q\|_{L^2[x, x+\delta]}. \tag{27}$$

Proof. Translate x to zero. Put $\alpha = 8N/\delta + 1$, $H(t) = e^{-\alpha t/2}Q(t)$. By (9), $\|H\|_{L^2(0,\infty)} \le \sqrt{2}\|Q\|_{L^2[0,\delta]}$. The stable roots $-\zeta_j$, $\zeta_j = \alpha/2 - z_j$, satisfy

$$b_j = \operatorname{Re}\zeta_j \le 5N/\delta, \quad S := \sum_j b_j \le 5N^2/\delta, \quad \max_j |\zeta_j| \le 6N/\delta, \quad \alpha/2 \le \tfrac{9}{2}N/\delta.$$

Expanding in Lemma 3.1 and using (14) gives

$$|H(0)| \le \sqrt{2S}\|H\|_2, \qquad |H'(0)| \le \sqrt{2S}(\max_j |\zeta_j| + 2S)\|H\|_2.$$

Since $Q'(0) = H'(0) + (\alpha/2)H(0)$,

$$|Q'(0)| \le \frac{41\sqrt{20}}{2}N^3\delta^{-3/2}\|Q\|_{L^2[0,\delta]} < 100N^3\delta^{-3/2}\|Q\|_{L^2[0,\delta]}.$$

Repeated roots are included directly. □

Proposition 6.2 (A global weighted derivative bound)**.** *Under Lemma 6.1, for $N \ge 4096$,*

$$\|(WQ)'\|_2 \le b_N\|WQ\|_2, \qquad b_N = KN^4, \qquad K = 101(100e^{118} + 112). \tag{28}$$

The constant is deliberately loose and independent of root separation.

Proof. Write $A = WQ$ and choose $\delta(x) = e^{-2(|x|+1)}$. For $y \in [x, x+\delta(x)]$, (18) gives $|\log W(y) - \log W(x)| \le 112\delta(x)e^{2(|x|+1)} = 112$. Thus (27) gives

$$|W(x)Q'(x)|^2 \le 100^2N^6e^{224}\delta(x)^{-3}\int_x^{x+\delta(x)} |A(y)|^2\,dy.$$

When $x \le y \le x + \delta(x)$, one has $|x - y| \le 1$ and $e^{-2}\delta(y) \le \delta(x) \le e^2\delta(y)$. For each y the set of such x has length at most $e^2\delta(y)$. Fubini gives

$$\|WQ'\|_2^2 \le 100^2N^6e^{224+8}\int \delta(y)^{-2}|A(y)|^2\,dy = 100^2N^6e^{236}\int e^{4|y|}|A(y)|^2\,dy.$$

Also $\|W'Q\|_2 \le 112\|e^{2|x|}A\|_2$. The moment bound (22) now gives

$$\|A'\|_2 \le (100e^{118}N^3 + 112)\sqrt{M_N^2 + 1}\|A\|_2.$$

Since $M_N \le 100N$ and $\sqrt{M_N^2+1} \le 101N$ for $N \ge 4096$, (28) follows. Integrability before normalization follows from the theta envelope, so no unknown derivative norm was used to estimate itself. □

7 Carrier separation and the complete Gamma comparison

For the unitary Fourier transform $\widehat{u}(\xi) = (2\pi)^{-1/2}\int u(x)e^{-i\xi x}\,dx$, Plancherel and the nonnegative difference integral give

$$t_\Gamma[u] = \int_{\mathbb{R}} d_\Gamma(\xi)|\widehat{u}(\xi)|^2\,d\xi, \qquad d_\Gamma(\xi) = 2\int_0^\infty \rho(s)(1 - \cos(\xi s))\,ds. \tag{29}$$

Since $(2s)^{-1} \le (1-e^{-2s})^{-1} \le 1+(2s)^{-1}$,

$$\tfrac{1}{2}\log(1+4\xi^2) \le d_\Gamma(\xi) \le 4+\tfrac{1}{2}\log(1+4\xi^2). \tag{30}$$

Here $\int_0^\infty e^{-s/2}(1-\cos(\xi s))\,ds/s = \frac{1}{2}\log(1+4\xi^2)$, by differentiation in ξ and integration back from zero. The remaining upper integral is $2\int_0^\infty e^{-s/2}(1-\cos(\xi s))\,ds \le 4$. No Gamma displacement or frequency is truncated.

Let Q have budget at most $N \ge 4096$, $|\operatorname{Re} z_j| \le 1/2$ and $0 \le \operatorname{Im} z_j \le L \le N$. Set

$$A = WQ, \qquad u_T(x) = \frac{e^{iTx}A(x) + e^{-iTx}A(-x)}{2}. \tag{31}$$

Only the positive envelope budget is N; the paired representation may have budget $2N$. Reflection is $A(-x)$, not $\overline{A(x)}$.

Proposition 7.1 (Complete whole packet estimate)**.** *Suppose $T \ge \max(1, 2b_N)$. Set*

$$h = \log T, \quad h_T = \tfrac{1}{2}\log(1+T^2), \quad \eta_{T,N} = 16b_N^2/T^2, \quad C_\Gamma = 4+\tfrac{1}{2}\log 37. \tag{32}$$

Then

$$|p[u_T]| \le P_N\|u_T\|_2^2, \tag{33}$$

$$h_T(1-\eta_{T,N})\|u_T\|_2^2 \le t_\Gamma[u_T] \le (h+C_\Gamma)\|u_T\|_2^2, \tag{34}$$

$$|q_0[u_T] - h\|u_T\|_2^2| \le (|c_\Gamma| + P_N + \max(C_\Gamma, \eta_{T,N}h_T))\,\|u_T\|_2^2. \tag{35}$$

The prime cap uses the positive envelope budget N, not $2N$.

Proof. The reflected carrier correlation is $I_T = \int e^{2iTx}A(x)\overline{A(-x)}\,dx$. The nonoscillatory factor has derivative L^1 norm at most $2\|A'\|_2\|A\|_2$ and vanishing endpoint terms. Integration by parts gives

$$|I_T| \le (b_N/T)\|A\|_2^2, \quad \|u_T\|_2^2 \ge \tfrac{1}{2}(1-b_N/T)\|A\|_2^2, \quad \|A\|_2 \le 2\|u_T\|_2. \tag{36}$$

Triangle inequality and reflection transfer (21) to

$$\|\mathbf{1}_{\{|x|>r\}}u_T\|_2 \le 2e^{-2e^{2r}}\|u_T\|_2 \quad (r \ge R_N).$$

The prime head is charged directly to $\|u_T\|_2^2$. For $n > \lceil M_N\rceil$, (26) now gives $|C_{u_T}(\log n)| \le 4e^{-2n}\|u_T\|_2^2$. The remaining series costs at most $8\sum_{n>\lceil M_N\rceil} e^{-2n}\|u_T\|_2^2 < \|u_T\|_2^2$, proving (33). The factor two in (36) affects only the exponentially small prime tail, not the head.

Plancherel and (28) imply

$$\|\mathbf{1}_{[-T/2,T/2]}\widehat{u_T}\|_2 \le (2b_N/T)\|A\|_2 \le (4b_N/T)\|u_T\|_2.$$

On the complementary Fourier region, (30) is at least h_T, proving the lower Gamma bound. Differentiation of (31) gives $\|u_T'\|_2 \le (T+b_N)\|A\|_2 \le 2(T+b_N)\|u_T\|_2$. For $u_T \ne 0$, Jensen's inequality gives

$$\frac{t_\Gamma[u_T]}{\|u_T\|_2^2} \le 4+\tfrac{1}{2}\log\left(1+4\frac{\|u_T'\|_2^2}{\|u_T\|_2^2}\right) \le 4+\tfrac{1}{2}\log(1+16(T+b_N)^2) \le h+C_\Gamma.$$

The last step uses $T \ge 1$ and $b_N/T \le 1/2$. Since $h_T \ge h$, the lower Gamma error from h is at most $\eta_{T,N}h_T$. Add the absolute scalar and prime costs to obtain (35). The zero function needs no division. □

8 The complete actual band and its zero count

The asymptotic conclusions need only the classical Riemann–von Mangoldt formula $N_\zeta(t) = F(t) + O(\log t)$, with multiplicities [4, 3]. To declare a concrete scalar upper budget without enumerating the zeros, we invoke the published explicit bound [3, Corollary 1.2]: for $t \geq e$,

$$|N_\zeta(t) - F(t)| \leq R(t), \quad F(t) = \frac{t}{2\pi}\log\frac{t}{2\pi e}, \quad R(t) = 0.1038\log t + 0.2573\log\log t + 9.3675. \tag{37}$$

Here N_ζ counts positive ordinates with multiplicity. The proof of this explicit refinement in [3] uses finite height numerical zero verification. We perform no such enumeration here; the asymptotic band conclusions do not depend on the particular numerical constants in this refinement. Endpoint conventions will not matter because we widen both endpoints. For $T \geq e+1$ put

$$\mathcal{M}(T,L) = F(T+L+1) - F(T-1) + R(T+L+1) + R(T-1). \tag{38}$$

Thus the full multiplicity count $M(T,L)$ in the closed band satisfies $M(T,L) \leq \mathcal{M}(T,L)$. Uniformly for $0 \leq L = o(T)$, the mean value theorem gives

$$M(T,L) \leq \frac{L+2}{2\pi}\log T + O\left(\log T + \frac{(L+2)^2}{T}\right) = O((L+1)\log T). \tag{39}$$

This is an upper bound, not a relative short interval counting asymptotic.

Every $u \in V_{T,L}$ has exactly the representation (31), with

$$Q(x) = \sum_{w:\ T\leq \operatorname{Re} w\leq T+L} \ \sum_{k<\nu(w)} c_{w,k}(ix)^k e^{i(w-T)x}. \tag{40}$$

Its positive envelope budget is at most $M(T,L)$ and its roots have imaginary parts in $[0,L]$ and real parts in $[-1/2,1/2]$. Differentiate both exponentials in the cosine to verify the identity; it is valid for complex w and complex $c_{w,k}$.

Proof of Theorem 1.1. Write $h = \log T$. Choose the common padded integer budget

$$n(T) = \lceil \max\{4096, h, L(T), \mathcal{M}(T, L(T))\} \rceil . \tag{41}$$

No zero enumeration is required. By (39) and (5),

$$n(T) = O((L(T)+1)h), \qquad n(T) \geq h, \qquad n(T) = O(h^2\sqrt{\log h}).$$

Consequently $b_{n(T)}/T \to 0$ and $h_T b_{n(T)}^2/T^2 \to 0$, so Proposition 7.1 applies eventually. The complete prime cap gives

$$\frac{P_{n(T)}}{h} = O\left(\frac{\sqrt{n(T)}}{h(\log n(T))^{1/4}}\right) = O\left(\sqrt{\frac{L(T)+1}{h\sqrt{\log h}}}\right) = o(1). \tag{42}$$

Now (35), divided by h, tends to zero uniformly over (40), proving (6). For $L = h(\log h)^{1/4}$ the prime contribution is $O((\log h)^{-1/8})$. □

Proof of Corollary 1.2. Fix $d > 0$ and set $L = dh\sqrt{\log h}$. The explicit upper budget (41) satisfies

$$n(T) = \left(\frac{d}{2\pi} + o(1)\right) h^2\sqrt{\log h}, \qquad \log n(T) = 2\log h + o(\log h).$$

This is an asymptotic for the declared upper budget using $\mathcal{M}$, not an unsupported relative asymptotic for the zeros actually present. It follows that $\limsup_{T\to\infty} P_{n(T)}/h \leq 120\sqrt{d}/(2^{3/4}\sqrt{\pi})$. The complete Gamma coefficient tends to one, and $c_\Gamma/h \to 0$. Subtracting the full absolute prime cap proves (7), whose lower bound is positive in the stated strict range of d. □

For any spanning list $v_1, \dots, v_m$, including a redundant list, let $G_{ij} = \langle v_j, v_i \rangle_2$ and $Q_{ij} = q_0(v_j, v_i)$. The uniform result is equivalently

$$(1 - \varepsilon_T) hG \preceq Q \preceq (1 + \varepsilon_T) hG.$$

No raw coefficient Gram eigenvalue, frequency gap or condition number is divided out. A represented zero vector lies in the kernel of both sides.

9 An exact inverse retaining Schur consequence

The whole band estimate controls screening on every subspace of that same band. The following version preserves the full prime error and its inverse response debit. On even finite theta exponential polynomials define

$$\begin{aligned} \mathcal{K}(u, v) &= \int_{\mathbb{R}} e^{-|s|/2} \int_{\mathbb{R}} u(x + s) \overline{v(x)} \, dx \, ds, \\ B &= t_\Gamma - c_\Gamma \langle \cdot, \cdot \rangle_2 + \mathcal{K}, \qquad E = q_0 - B = -p - \mathcal{K}. \end{aligned} \tag{43}$$

The Fourier multiplier of $\mathcal{K}$ is $(\xi^2 + 1/4)^{-1}$, so $0 \preceq \mathcal{K} \preceq 4G$. On the band in Theorem 1.1, eventually

$$(h - C_B) G \preceq B \preceq (h + C_B) G, \qquad C_B = C_\Gamma + |c_\Gamma| + 5, \tag{44}$$

by (34) with $\eta_{T,n} h_T \le 1$. This gives a positive reference on the entire represented band without a separately assumed sign of q_0. The full arithmetic cap gives

$$-(P_n + 4) G \preceq E \preceq P_n G. \tag{45}$$

Corollary 9.1 (Signed screened determinant)**.** *Let a finite represented union $R + V$ lie in one band satisfying (44), with $h > C_B$. Set $r_T = (P_n + 4)/(h - C_B) < 1$. Choose residual sources spanning R after B orthogonal projection off V, and let H be their B Gram. Let Q_{VV} be the actual complete q_0 matrix on represented support of V. It is positive definite there. Exact screening gives*

$$S = H + E_{RR} - E_{RV} Q_{VV}^{-1} E_{VR}, \qquad (1 - r_T) H \preceq S \preceq (1 + r_T) H. \tag{46}$$

For two residual sources, put $F = E_{RR} - E_{RV} Q_{VV}^{-1} E_{VR}$. Then

$$\operatorname{tr}(\operatorname{adj}(H) F) + \det F \ge (-2 r_T + r_T^2) \det H. \tag{47}$$

Under (5), $r_T \to 0$. All inverses act on represented support, not redundant coordinates.

Proof. From (44) and (45), $(1 - r_T) B \preceq q_0 \preceq (1 + r_T) B$ on the complete union. Thus Q_{VV} is positive on represented support. For residual $x \in R$, minimization over $v \in V$ gives

$$(1 - r_T) B[x] \le \inf_{v \in V} q_0[x + v] \le (1 + r_T) B[x],$$

since $B(x, V) = 0$. Completing the square in the actual V block gives (46), including the nonnegative inverse response debit. When H is positive definite, normalization by $H^{-1/2}$ implies $\det S \ge (1 - r_T)^2 \det H$. Use the exact two by two identity $\det(H + F) = \det H + \operatorname{tr}(\operatorname{adj}(H) F) + \det F$. When H is singular, (46) gives the same kernel for H and S, so both determinants vanish and the identity still proves (47). Finally (42) gives $r_T \to 0$. □

The debit has not been replaced by an ordinary Gram inverse, bounded entry by entry, or omitted. Control comes from the joint all coefficients estimate on the complete union. This is not a gluing argument from two neighboring positive bands. A union wider than the proved range is not covered.

10 Verification and remaining limitations

The estimates proved here are analytic. We perform no numerical theta integration, zero enumeration or matrix sign computation, and we do not use the prime number theorem. The explicit zero counting theorem (37) is invoked as published, including its finite height numerical verification dependency. The qualitative band conclusions need only the classical asymptotic zero count, so no claim that the complete dependency chain of the explicit refinement is nonnumerical is needed. The constants 4096, 100, 12 and the boundary coefficient are conservative consequences of the displayed inequalities. The very large K in (28) is explicit but is not advertised as an efficient numerical height bound. For specified T, L, (41), the carrier condition $T \geq 2Kn(T)^4$ and (35) provide a scalar sufficient test. No final numerical starting height is evaluated here.

The accompanying script checks exact scalar inequalities, the basis endpoint formulas in simple configurations, and the two by two determinant algebra. Floating point sample checks are labelled as illustrations, not substitutes for whole ray proofs. The editable source, PDF and reproducibility files are intended for https://github.com/serabing-hash/theta-weighted-certificate/tree/main/publications/growing-zero-jet-bands-2026-11-03. This source includes all arguments except the classical zeta strip location and the cited zero counting theorem. Compilation uses standard LaTeX and requires no private data or computer algebra system.

Three restrictions remain essential. The prime estimate controls bounded prefix complexity, not arbitrary infinite exponential sums. The actual zero theorem covers an entire single contiguous band, not the mixed terms in an arbitrary covering of all positive ordinates. Ordinary L^2 and reciprocal theta weighted norms cannot be interchanged with a budget independent constant. The Schur corollary does not remove these restrictions. The unresolved global sign, and any deduction of the Riemann hypothesis through that sign, remains outside this result.

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